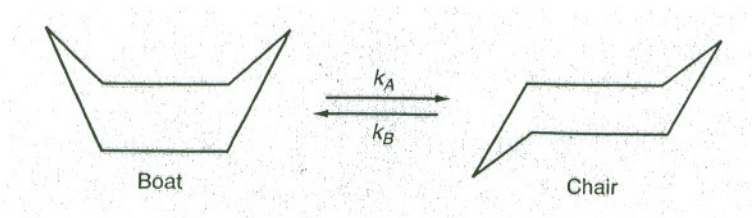
Physical Cume

September 29, 2006

(1) Consider the interconversion of the "boat" and "chair" conformations of cyclohexane:



The reaction is first order in each direction, with an equilibrium constant of 1×10^4. The activation energy for the conversion of the "chair" conformer to the boat conformer is 42 kJ/mol. Assuming an Arrhenius preexponential factor of 1×10^{12} s^{-1}., what is the expected observed reaction rate constant (k_A) at 298 K if one were to initiate this reaction starting with only the "boat" conformer?

(2)Bananas are somewhat radioactive due to the presence of substantial amounts of potassium. Potassium-40 decays by two different paths:

$$^{40}_{19}K \rightarrow ^{40}_{20}Ca + \beta \quad (89.3\%)$$
$$^{40}_{19}K \rightarrow ^{40}_{18}Ar + \beta^+ \quad (10.7\%)$$

The half-life for potassium decay is 1.3×10^9 years. Determine the rate constants for the individual channels in units of $year^{-1}$.

(3) Consider the following reaction of methane with molecular chlorine:

$$CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$$

Experimental studies have shown that the rate law for the production of HCl(g) is one-half order with respect to Cl_2. Determine if the following mechanism is consistent with this behavior with respect to Cl_2. **·X** is a free radical species.

$$Cl_2 \xrightarrow{k_1} 2Cl\bullet$$
$$Cl\bullet + CH_4 \xrightarrow{k_2} HCl + CH_3\bullet$$
$$CH_3\bullet + Cl_2 \xrightarrow{k_3} CH_3Cl + Cl\bullet$$
$$Cl\bullet + Cl\bullet \xrightarrow{k_4} Cl_2$$